Polynomials and Rational Functions

Course: Pre-Calculus-I

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Outline

1 Polynomials

2 Rational Functions

3 Class Practice

What is a Polynomial?

Definition

A polynomial in x is a function of the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where $n \geq 0$ is an integer, $a_n \neq 0$, and each $a_i \in \mathbb{R}$.

- a_n is the leading coefficient, a_0 is the constant term.
- n is the degree of the polynomial.
- The domain of every polynomial is $\mathbb{R} = (-\infty, \infty)$.

Note

Polynomials are continuous and smooth everywhere — no breaks, holes, or sharp corners.

Examples and Non-Examples

Polynomials

- $P(x) = 3x^4 - 2x^2 + 7$ (degree 4)
- $Q(x) = -5x + 1$ (degree 1, linear)
- $R(x) = 2$ (degree 0, constant)
- $S(x) = x^3 - x$ (degree 3, cubic)

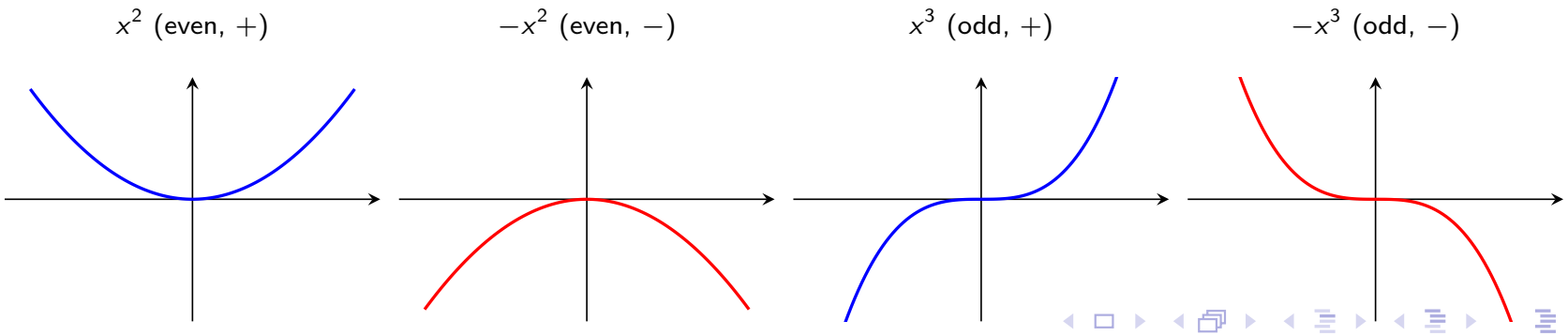
Not Polynomials

- $\frac{1}{x}$ (negative exponent)
- $\sqrt{x} = x^{1/2}$ (non-integer exponent)
- $|x|$ (not a single power sum)
- 2^x (variable in exponent)

End Behaviour of Polynomials

The end behaviour depends on the degree n and the leading coefficient a_n.

Degree	Leading coeff.	End behaviour
even	$a_n > 0$	up on both sides
even	$a_n < 0$	down on both sides
odd	$a_n > 0$	down left, up right
odd	$a_n < 0$	up left, down right



Roots and Factored Form

Factor Theorem

$x = c$ is a root of P (i.e. $P(c) = 0$) iff $(x - c)$ is a factor of $P(x)$.

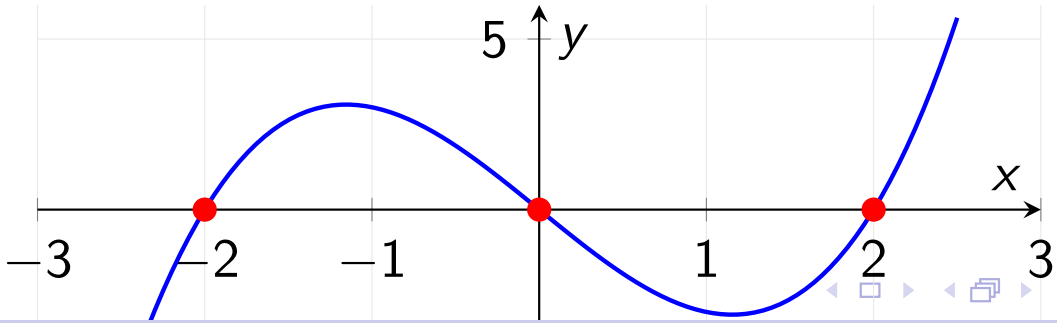
A polynomial of degree n has at most n real roots and can be written as

$$P(x) = a_n(x - r_1)(x - r_2)\cdots(x - r_n)$$

over $\mathbb{C}$ (counting multiplicity).

Example: $P(x) = x^3 - 4x = x(x - 2)(x + 2)$.

- Roots: $x = 0,\ 2,\ -2$.
- Degree 3, leading coeff. $1 \Rightarrow$ odd, up-right end behaviour.



What is a Rational Function?

Definition

A rational function is a quotient of two polynomials:

$$f(x) = \frac{P(x)}{Q(x)}, \qquad Q(x) \neq 0.$$

- Domain: all real x with $Q(x) \neq 0$.
- Points where $Q(x) = 0$ (but $P(x) \neq 0$) give vertical asymptotes.
- End behaviour is governed by the degrees of P and Q.

Note

A rational function can have holes where both P and Q vanish at the same x.

Examples of Rational Functions

Examples

- $f(x) = \dfrac{1}{x}$ domain $\mathbb{R} \setminus \{0\}$
- $g(x) = \dfrac{x+1}{x-2}$ domain $\mathbb{R} \setminus \{2\}$
- $h(x) = \dfrac{x^2-1}{x^2-4} = \dfrac{(x-1)(x+1)}{(x-2)(x+2)}$

A Hole Example

$$k(x) = \frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1} = x+1, \quad x \neq 1.$$

The graph is the line $y = x+1$ with a hole at $(1,2)$.

Asymptotes of Rational Functions

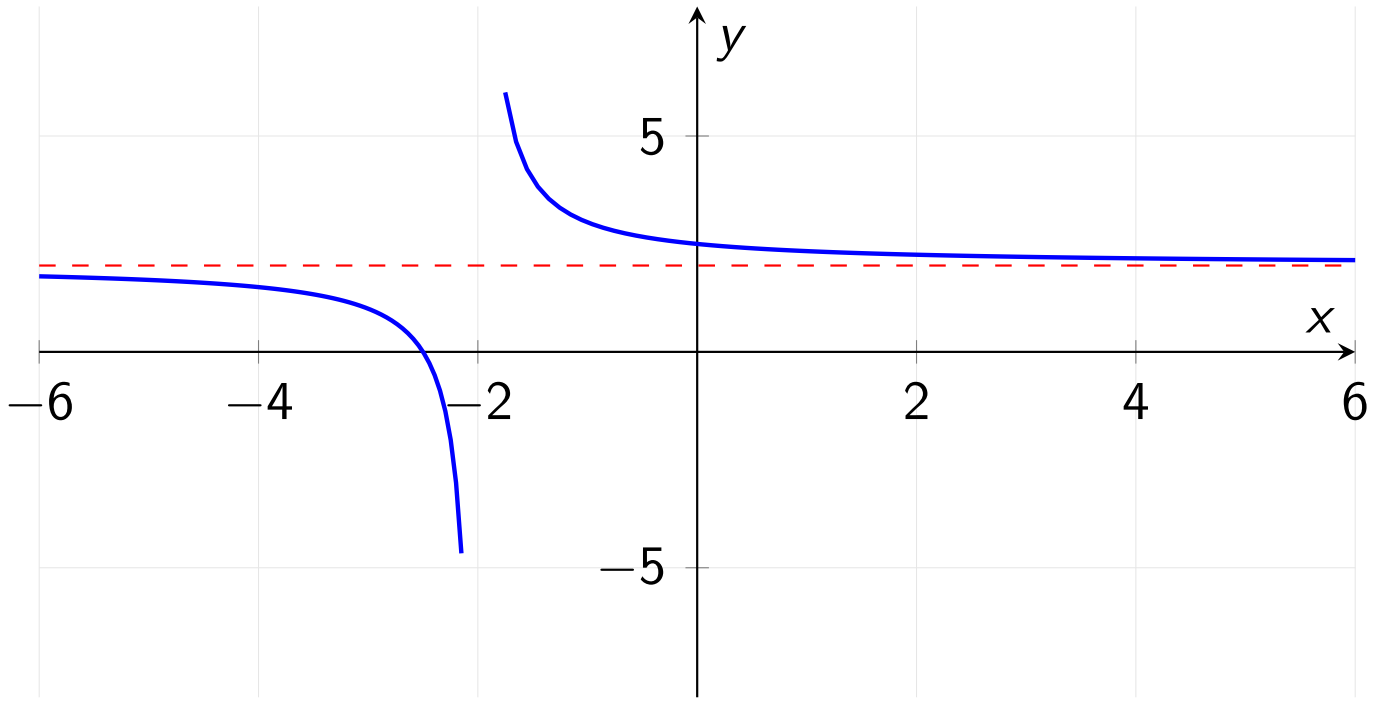
Let $f(x) = \dfrac{P(x)}{Q(x)}$ with degrees $n = \deg P$, $m = \deg Q$.

Degree comparison	**Horizontal asymptote**
$n < m$	$y = 0$
$n = m$	$y = \dfrac{\text{lead coeff of } P}{\text{lead coeff of } Q}$
$n > m$	none (oblique if $n = m + 1$)

Example: $f(x) = \dfrac{3x^2 + 1}{x^2 - 4}$.

- Vertical asymptotes: $x = \pm 2$.
- Horizontal asymptote: $y = 3/1 = 3$ (equal degrees).

Graph of a Rational Function



$f(x) = \dfrac{2x+5}{x+2}$ has vertical asymptote $x = -2$ and horizontal asymptote $y = 2$.

Worked Example: Analyse $f(x) = \dfrac{x+1}{x-2}$

Step 1 (Domain): $x - 2 \neq 0 \Rightarrow x \neq 2$. Domain $= \mathbb{R} \setminus \{2\}$.
Step 2 (Vertical asymptote): $x = 2$ (numerator $= 3 \neq 0$).
Step 3 (Horizontal asymptote): equal degrees $\Rightarrow y = 1/1 = 1$.
Step 4 (Intercepts):

- y-intercept: $f(0) = -1/2$.
- x-intercept: $x + 1 = 0 \Rightarrow x = -1$.

Step 5 (Sign chart): test $x = -2, 0, 3$:

Interval	$x+1$	$x-2$	$f(x)$
$(-\infty, -1)$	$-$	$-$	$+$
$(-1, 2)$	$+$	$-$	$-$
$(2, \infty)$	$+$	$+$	$+$

Class Practice Questions

P1. State the degree, leading coefficient, and end behaviour of

$$P(x) = -2x^5 + 3x^3 - x + 4.$$

P2. Factor completely and find all roots: $P(x) = x^3 - 9x$.

P3. Find the domain, vertical asymptote(s), and horizontal asymptote of

$$f(x) = \frac{2x-1}{x+3} \text{ and } g(x) = \frac{x^2-4}{x^2-1}$$

P4. Identify any hole in $h(x) = \dfrac{x^2-9}{x-3}$ and state its coordinates.

P5. Consider $f(x) = \dfrac{x^2+x-6}{x^2-x-2}$.

(a) Factor numerator and denominator completely.

(b) State the domain and identify any holes.

(c) Find all vertical and horizontal asymptotes.

(d) Determine the x- and y-intercepts, then sketch the graph.

Class Practice Questions

P6. Let $f(x) = \dfrac{x^2 - 2x - 3}{x - k}$, where k is a real constant.

(a) For which value(s) of k does the graph have a hole? Give the hole's coordinates.

(b) For which value(s) of k does the graph have a vertical asymptote? State it.

(c) Find the oblique asymptote (if any) when $k = 1$, and describe the end behaviour of f as $x \to \pm\infty$.

P7. Let $f(x) = \dfrac{2x^2 - 3x - 2}{x^2 - 4}$.

(a) Factor the numerator and denominator completely, then simplify.

(b) State the domain, and identify any holes and their coordinates.

(c) Find all vertical and horizontal asymptotes.

(d) Solve $f(x) = 0$ and $f(x) < 0$ using a sign chart.

P8. A rational function has vertical asymptotes at $x = -1$ and $x = 3$, a horizontal asymptote $y = 2$, and an x-intercept at $x = 1$.

(a) Write a possible formula for $f(x)$ of the form $\dfrac{a(x - r)}{}$ and

Class Practice Questions

P8. A rational function has vertical asymptotes at $x = -1$ and $x = 3$, a horizontal asymptote $y = 2$, and an x-intercept at $x = 1$.

(a) Write a possible formula for $f(x)$ of the form $\dfrac{a(x-r)}{(x+1)(x-3)}$ and determine a from the horizontal asymptote.

(b) State the domain and the y-intercept of your function.

(c) Sketch the graph, showing the sign of f on each interval determined by the asymptotes and intercept.